RAG Assignment: Baseline vs. Re-Ranking – Report

Roll No: 2023101025 **GitHub:** `https://github.com/Jaswanth2534/RAG-LMA-Assignment`

Document: *Twilight Imperium 4th Ed. Living Rules Reference v2.0* (44 pages, 304 chunks). Bi-Encoder: `all-MiniLM-L6-v2` + FAISS. Cross-Encoder: `ms-marco-MiniLM-L-6-v2`. LLM: `gemini-3.5-flash-lite`, temperature 0, same prompt for both pipelines.

1. Architecture Comparison

Bi-Encoder. The query and each chunk are turned into vectors *separately*. Relationship btw them is the cosine similarity between the two vectors. Chunk vectors are computed once, before any query, so search is very fast. The weakness is that a whole chunk is squeezed into one vector, so small but important words such as "not", "other players'" or "except" have little effect on the score. The query and chunk never "see" each other.

Cross-Encoder. The query and one chunk are given to the model *together* as one input. Self-attention lets every query word look at every chunk word, and the model outputs one relevance score. This makes it much better at noticing negations, exceptions and synonyms. The cost is that nothing can be pre-computed: the model must run once for every (query, chunk) pair at query time.

2. Qualitative & Rank Shift Analysis

Rank shifts. For each query, the "gold" chunk is the chunk that contains the rule which answers it. In all six queries the gold chunk was *outside* the baseline Top 3 and the Cross-Encoder moved it *into* the Top 3.

Type	Query	Gold chunk (rule)	Rank: Bi → Cross
Failure query 1	Can I activate a system where I already have a command token?	Chunk 48 (5.2)	#8 → #1
Failure query 2	Can I spend resources from a planet I already tapped this round?	Chunks 156 / 155 (64.9)	#4 / #5 → #2 / #3
Keyword trap	Do players who are not in the battle get to use combat abilities?	Chunk 159 (62.1)	#5 → #1
Exception	If my space dock is blockaded, can I still produce anything there?	Chunk 58 (14.1)	#4 → #1
Negation	Which hits can Sustain Damage not cancel?	Chunk 210 (87.5)	#4 → #1
Buried (rank 4–10)	Can I still make trades on the turn that I pass?	Chunk 40 (3.3a)	#6 → #2

Detailed example – Failure query 1: *"Can I activate a system where I already have a command token?"*
The correct answer is **No**. Rule 5.2 (chunk 48): *"A player cannot activate a system that already contains one of their command tokens."* The table shows the baseline Top 3, the chunk that the Cross-Encoder promoted, and the gold chunk.

Chunk	What it says	Bi rank (score)	Cross rank (score)
244	One sentence about putting a gained command token into a pool on the command sheet	#1 (0.502)	#6 (−2.46)
213	Rule 89.1b: *"Other players' command tokens do not prevent a player from activating a system"*	#2 (0.469)	#5 (0.11)
69	What happens when a player has no command tokens left in reinforcements	#3 (0.455)	#3 (1.78)
94	Diplomacy card: no command token is placed if one is already in the chosen system	#4 (0.421)	#2 (3.01)
48	**Rule 5.2: a player cannot activate a system that already contains one of their command tokens**	**#8 (0.362)**	**#1 (5.01)**

- **Why the baseline failed:** chunks 244, 213 and 69 share many words with the query ("activate", "system", "command token"), so their vectors are close to the query vector. Chunk 213 is a trap: it has almost the same words but talks about *other players'* tokens and says they do *not* prevent activation. One vector per chunk cannot keep "other players" vs. "my own" and "do not prevent" vs. "cannot" apart, so the gold chunk fell to #8.
- **Answer 1 (baseline):** *"Yes, other players' command tokens do not prevent a player from activating a system [Chunk 213]…"* This is a **misstated rule**: the model took the rule about *other* players' tokens and applied it to the player's own token, because rule 5.2 was not in its context.
- **Answer 2 (re-ranked):** *"No, you cannot activate a system where you already have one of your command tokens [Chunk 48]."* This is correct. The Cross-Encoder reads the query and the chunk together, so it matched "already have a command token" with "already contains one of their command tokens" and moved chunk 48 from #8 to #1.

Detailed example – Failure query 2: *"Can I spend resources from a planet I already tapped this round?"*
The correct answer is **No**. Rule 64.9 (in chunks 155 and 156, which overlap): *"A player cannot spend an exhausted planet's resources or influence."*

Chunk	What it says	Bi rank (score)	Cross rank (score)
181	Rule 75: a planet's resources value, and *"a player spends a planet's resources by exhausting its card"*	#1 (0.394)	#1 (−1.28)
180	Mostly relics and dice rerolls; only the heading of the Resources section	#2 (0.382)	#5 (−7.29)
85	Unit cost: how many resources must be spent to produce a unit	#3 (0.368)	#4 (−6.70)
156	**Rule 64.8–64.9: readied planets can be spent, exhausted planets cannot** (+ Planetary Shield text)	**#4 (0.362)**	**#2** (−1.75)
155	**Planet cards are readied (faceup) or exhausted (facedown); rule 64.9**	**#5 (0.350)**	**#3** (−4.83)

- **Why the baseline failed:** this is a synonym problem. The query says "tapped", but the rulebook never uses that word; it says "exhausted". The bi-encoder ranked chunks by shared words like "spend", "resources" and "planet", so chunks about spending resources and unit cost won. The gold chunks also contain unrelated text (Planetary Shield, legendary planets), which weakens their single vector further.
- **Answer 1 (baseline):** it explained that *"a player spends a planet's resources by exhausting its card [Chunk 181]"*, then said *"the specific rules for whether an already exhausted/tapped planet can be used again in the same round are not detailed in the provided chunks."* This **omits the essential clause** (64.9), so the user gets no answer.
- **Answer 2 (re-ranked):** *"You cannot spend resources from a planet you have already tapped (exhausted)… A player cannot spend an exhausted planet's resources or influence [Chunk 156]."* This is correct. Reading the query and chunk together, the Cross-Encoder linked "tapped" to "exhausted" and moved both gold chunks into the Top 3 (#4 → #2 and #5 → #3). All Cross-Encoder scores here are negative, but only their order matters: the gold chunks scored higher than the two off-topic chunks (180 and 85), which dropped out.

Answer 1 vs. Answer 2 for all queries. The prompt asks the LLM to answer based on the given chunks and to cite them.

Query	Answer 1 (baseline)	Answer 2 (re-ranked)
Failure query 1	**Misstated rule**: said "yes", misled by chunk 213 (see above).	Correct
Failure query 2	**Omitted clause**: explained how resources are spent, but missed rule 64.9 (see above).	Correct
Keyword trap	**Omitted clause**: said the document has no such rule.	Correct
Exception	**Misstated rule**: said there is no restriction; missed "cannot produce ships, can still produce ground forces".	Correct
Negation	**Omitted clause**: gave a nearby rule (87.3), missed 87.5.	Correct
Buried	**Omitted clause**: said it could not confirm either way.	Correct

Answer 2 was correct in all six queries. I did not find a pure hallucination: every claim in Answer 1 came from one of its three chunks. The wrong answers came from the model drawing the wrong conclusion from nearby, partly related chunks, because the critical chunk sat outside the Top 3.

3. Latency Profiling

Measured on a Colab T4 GPU, 20 runs per query after a warm-up run. LLM time is not included.

Query	P1 retrieval (ms)	Re-rank only (ms)	P2 total (ms)	Slowdown
Example: produce ships in a system with other players' ships	14.91 ± 6.62	43.55	58.46	3.92×
Failure 1: activate system with own command token	12.15 ± 5.60	36.40	48.56	4.00×
Failure 2: spend resources from a tapped planet	11.59 ± 3.12	33.82	45.41	3.92×
Keyword trap: players not in the battle	11.23 ± 2.18	31.62	42.85	3.82×
Exception: blockaded space dock	10.68 ± 1.41	32.20	42.88	4.01×
Negation: what Sustain Damage cannot cancel	12.29 ± 4.51	33.41	45.70	3.72×
Buried: trades on the turn I pass	12.02 ± 9.23	36.43	48.45	4.03×
Average	**12.12**	**35.35**	**47.47**	**3.92×**

Pipeline 2 total = bi-encoder retrieval + cross-encoder re-ranking of 10 pairs. Re-ranking makes retrieval about 4× slower, but it only adds about 35 ms.

4. Scaling Question

A Cross-Encoder has to run the full model once for *every* (query, chunk) pair, and nothing can be stored in advance because the score depends on the query. In my tests 10 pairs took about 35 ms (about 3.5 ms per pair). Scoring all 304 chunks would take about 1 s per query, and when we try to extend this millions of chunks it takes lot time and compute making it not scalable.

A Bi-Encoder avoids this: chunk vectors are computed once, offline, and each query needs only one encoding plus a fast vector search, which scales to millions of chunks. So the best design is two stages: the Bi-Encoder cheaply narrows the whole collection down to a few candidates (Top 10), and the Cross-Encoder spends its expensive, accurate scoring only on those few.